

• Deepseek V3 variation -

Auxiliary loss free balancing:

set-up a per-expert bias (make it more likely to get tokens) and use online learning.

⇒ the system-wide to train MoEs

↓
MoEs parallelize nicely - Each FFM can fit in a device
↳ enables additional kinds of parallelism.

• MoE routing allows for parallelism, but also complexity

issues of MoE: training instability / fine-tuning hardship
over-fitting on smaller data

upcycling ⇒ use a pre-trained LM to initialize MoE
Deepseek

• fine-grained, shared MoE for Deepseek V3

new: sigmoid + softmax topK + topM / MLHA (latent attention)

MoE: load-balancing $\rightarrow$ expert router.

MoE routing allows for parallelism, but also some complexities

Megablocks: open MoE, library

MoE issue: training loss stability

Fine-tuning = sparse MoEs can overfit on smaller fine-tuning data.

solutions $\left\{ \begin{array}{l} \text{fine-tune non-MoE MLPs} \\ \text{use lots of data: 1.4T} \end{array} \right.$

Qwen MoE: initialized from Qwen 1.8B model $\text{top-k}=4$, 60 experts w/4 shared

Deepseek MoE: similar structure, but one of the first upcycling success

Deepseek Architecture: $V_1 - V_2 - V_3$

$V_1: \rightarrow (16B - 28 \text{ active})$
shared(2) + fine-grained(64/4) experts

standard, top-k routing
(tokens selection)

standard aux-loss balancing
(expert-diverse)

$V_2: (286B - 21 \text{ active})$

shared(2) + fine-grained(160/16) experts, 6 active